Homework 03: Solutions

REAL VARIABLES I

Fall 2026

Solutions for Homework 03.

1. (When a generated outer measure extends its cost)

1.a

Remark (Solution). By definition,

$$\ell^*(E) = \inf\Big\{\sum_{n\in\mathbb{N}} \ell(A_n) : A_n \in \mathcal{A},\ E \subseteq \bigcup_n A_n\Big\}, \qquad \inf\emptyset \coloneqq \infty.$$

We show that $\ell^*(A) = \ell(A)$ for every $A \in \mathcal{A}$ if and only if ℓ is σ-subadditive.
Step 1: $\ell^*(A) \leq \ell(A)$ **for every** $A \in \mathcal{A}$**.** The sequence $A_0 \coloneqq A$, $A_n \coloneqq \emptyset$ for $n \geq 1$, consists of sets in $\mathcal{A}$ and satisfies $A \subseteq \bigcup_n A_n$. Therefore, according to the definition of $\ell^*(A)$ as an infimum over such sequences, it holds that $\ell^*(A) \leq \sum_n \ell(A_n)$. Since $\ell(\emptyset) = 0$, this gives $\ell^*(A) \leq \ell(A) + \sum_{n\geq 1} \ell(\emptyset) = \ell(A)$.
Step 2 ($\Leftarrow$). Let $A \in \mathcal{A}$. We show that $\ell(A) \leq \sum_n \ell(A_n)$ for every sequence (A_n) in $\mathcal{A}$ with $A \subseteq \bigcup_n A_n$, that is, that $\ell(A)$ is a lower bound of the set whose infimum is $\ell^*(A)$. Let (A_n) be such a sequence. The hypothesis that ℓ is σ-subadditive states that $\ell(B) \leq \sum_n \ell(B_n)$ whenever $B, B_n \in \mathcal{A}$ and $B \subseteq \bigcup_n B_n$. Taking $B = A$ and $B_n = A_n$ gives $\ell(A) \leq \sum_n \ell(A_n)$. Since the infimum is the greatest lower bound, it holds that $\ell(A) \leq \ell^*(A)$. Together with Step 1, $\ell^*(A) = \ell(A)$.
Step 3 ($\Rightarrow$). We show that $\ell(A) \leq \sum_n \ell(A_n)$ whenever $A, A_n \in \mathcal{A}$ and $A \subseteq \bigcup_n A_n$. The sequence (A_n) consists of sets in $\mathcal{A}$ and satisfies $A \subseteq \bigcup_n A_n$. Therefore, according to the definition of $\ell^*(A)$ as an infimum over such sequences, it holds that $\ell^*(A) \leq \sum_n \ell(A_n)$. By the hypothesis of this direction, $\ell^*(B) = \ell(B)$ for every $B \in \mathcal{A}$, and $A \in \mathcal{A}$, so $\ell(A) = \ell^*(A)$. Hence $\ell(A) \leq \sum_n \ell(A_n)$. Since A and (A_n) were arbitrary, ℓ is σ-subadditive.

2. (Open approximation of Lebesgue outer measure)

2.a

Remark (Solution). Fix $E \subseteq \mathbb{R}$ and $\epsilon > 0$. We construct an open $A \supseteq E$ with $\mathcal{L}^*(A) \leq \mathcal{L}^*(E) + \epsilon$.
If $\mathcal{L}^*(E) = \infty$, take $A \coloneqq \mathbb{R}$. This set is open and contains E. Moreover, $\mathcal{L}^*(A) \leq \infty = \mathcal{L}^*(E) + \epsilon$.
If $\mathcal{L}^*(E) < \infty$, then $\mathcal{L}^*(E) + \epsilon$ is larger than the infimum $\mathcal{L}^*(E)$, so it is not a lower bound of the set of sums of lengths of countable covers of E by open intervals. Hence there are open intervals (a_j, b_j), $j \in \mathbb{N}$, with $E \subseteq \bigcup_j (a_j, b_j)$ and $\sum_j (b_j - a_j) < \mathcal{L}^*(E) + \epsilon$. Put $A \coloneqq \bigcup_j (a_j, b_j)$. It is open as a union of open intervals, and $E \subseteq A$. The intervals (a_j, b_j) are open intervals with $A = \bigcup_j (a_j, b_j)$. Therefore, according to the definition of $\mathcal{L}^*(A)$ as an infimum over such covers, it holds that $\mathcal{L}^*(A) \leq \sum_j (b_j - a_j)$. Hence $\mathcal{L}^*(A) < \mathcal{L}^*(E) + \epsilon$.

3. (Induced σ-algebras)

3.a

Remark (Solution). Let $\mathcal{F}_1 \coloneqq \{A \in \mathcal{E} : A \subseteq Y\}$ and $\mathcal{F}_2 \coloneqq \{A \cap Y : A \in \mathcal{E}\}$. We show that $\mathcal{F}_1 = \mathcal{F}_2$ and that $\mathcal{F}_1$ is a σ-algebra on Y.
$\mathcal{F}_1 = \mathcal{F}_2$. The hypotheses are that $\mathcal{E}$ is a σ-algebra on X and $Y \in \mathcal{E}$. Since $\mathcal{E}$ is closed under complements and countable unions, it is closed under finite intersections: $A \cap B = X \setminus ((X \setminus A) \cup (X \setminus B))$.
($\mathcal{F}_1 \subseteq \mathcal{F}_2$) If $A \in \mathcal{F}_1$, then $A \in \mathcal{E}$ and $A \subseteq Y$. Hence $A = A \cap Y$ with $A \in \mathcal{E}$, so $A \in \mathcal{F}_2$.
($\mathcal{F}_2 \subseteq \mathcal{F}_1$) If $B = A \cap Y$ with $A \in \mathcal{E}$, then $B \in \mathcal{E}$ because $A, Y \in \mathcal{E}$ and $\mathcal{E}$ is closed under finite intersections. Also $B \subseteq Y$. Hence $B \in \mathcal{F}_1$.
$\mathcal{F}_1$ is a σ-algebra on Y.

- $Y \in \mathcal{E}$ and $Y \subseteq Y$, so $Y \in \mathcal{F}_1$.
- If $A \in \mathcal{F}_1$, then $A \in \mathcal{E}$ and $A \subseteq Y$. Thus $X \setminus A \in \mathcal{E}$, so $Y \setminus A = Y \cap (X \setminus A) \in \mathcal{E}$ by closure under finite intersections. Also $Y \setminus A \subseteq Y$. Hence $Y \setminus A \in \mathcal{F}_1$.
- If $A_n \in \mathcal{F}_1$ for $n \in \mathbb{N}$, then $A_n \in \mathcal{E}$ and $A_n \subseteq Y$ for every n. Thus $\bigcup_n A_n \in \mathcal{E}$ and $\bigcup_n A_n \subseteq Y$. Hence $\bigcup_n A_n \in \mathcal{F}_1$.

3.b

Remark (Solution). Setting. Let $\mathcal{T}_X$ and $\mathcal{T}_E$ be the families of open sets of X and of E, where E carries the restricted metric. The open balls of E are $B_E(e, r) = B_X(e, r) \cap E$. Let $\mathcal{B}(X)|_E \coloneqq \{A \cap E : A \in \mathcal{B}(X)\}$ be the trace of $\mathcal{B}(X)$ on E. We show that $\mathcal{B}(E) = \mathcal{B}(X)|_E$.
Plan. The proof has four parts.

1. The open sets of E are exactly the traces $U \cap E$ of open sets U of X (a fact from topology, cited).
2. The trace $\mathcal{B}(X)|_E$ is a σ-algebra on E.
3. From parts 1 and 2 we obtain $\mathcal{B}(E) \subseteq \mathcal{B}(X)|_E$.
4. By a σ-algebra on X built from $\mathcal{B}(E)$, we obtain $\mathcal{B}(X)|_E \subseteq \mathcal{B}(E)$.

Part 1: a fact from topology. The topology of the metric subspace E is the subspace topology of $E \subseteq X$. Therefore, according to the definition of the subspace topology, it holds that

$$\mathcal{T}_E = \{U \cap E : U \in \mathcal{T}_X\}.$$

This is a standard fact of topology and is used without proof.
Part 2: $\mathcal{B}(X)|_E$ is a σ-algebra on E. Each of the three properties of a σ-algebra on E follows from the same property of $\mathcal{B}(X)$ and one identity.

- $E \in \mathcal{B}(X)|_E$, because $E = X \cap E$ and $X \in \mathcal{B}(X)$.
- Closure under complements in E. Let $A \in \mathcal{B}(X)$. Then $E \setminus (A \cap E) = (X \setminus A) \cap E$, and $X \setminus A \in \mathcal{B}(X)$. Hence $E \setminus (A \cap E) \in \mathcal{B}(X)|_E$.
- Closure under countable unions. Let $A_n \in \mathcal{B}(X)$ for $n \in \mathbb{N}$. Then $\bigcup_n (A_n \cap E) = (\bigcup_n A_n) \cap E$, and $\bigcup_n A_n \in \mathcal{B}(X)$. Hence $\bigcup_n (A_n \cap E) \in \mathcal{B}(X)|_E$.

Part 3: $\mathcal{B}(E) \subseteq \mathcal{B}(X)|_E$. Let $V \in \mathcal{T}_E$. By Part 1, $V = U \cap E$ for some $U \in \mathcal{T}_X$, and $U \in \mathcal{B}(X)$ because open sets are Borel. Hence $V \in \mathcal{B}(X)|_E$, which shows $\mathcal{T}_E \subseteq \mathcal{B}(X)|_E$. By Part 2, $\mathcal{B}(X)|_E$ is a σ-algebra on E. Since $\mathcal{B}(E)$ is the smallest σ-algebra on E containing $\mathcal{T}_E$, it holds that $\mathcal{B}(E) \subseteq \mathcal{B}(X)|_E$.
Part 4: $\mathcal{B}(X)|_E \subseteq \mathcal{B}(E)$. Let $\mathcal{G} \coloneqq \{A \subseteq X : A \cap E \in \mathcal{B}(E)\}$. We show that $\mathcal{G}$ is a σ-algebra on X containing $\mathcal{T}_X$.

- $X \in \mathcal{G}$, because $X \cap E = E \in \mathcal{B}(E)$.
- Closure under complements. Let $A \in \mathcal{G}$. Then $(X \setminus A) \cap E = E \setminus (A \cap E)$, which lies in $\mathcal{B}(E)$ because $A \cap E \in \mathcal{B}(E)$. Hence $X \setminus A \in \mathcal{G}$.
- Closure under countable unions. Let $A_n \in \mathcal{G}$ for $n \in \mathbb{N}$. Then $(\bigcup_n A_n) \cap E = \bigcup_n (A_n \cap E)$, which lies in $\mathcal{B}(E)$. Hence $\bigcup_n A_n \in \mathcal{G}$.
- $\mathcal{T}_X \subseteq \mathcal{G}$. Let $U \in \mathcal{T}_X$. By Part 1, $U \cap E \in \mathcal{T}_E \subseteq \mathcal{B}(E)$, so $U \in \mathcal{G}$.

Since $\mathcal{B}(X)$ is the smallest σ-algebra on X containing $\mathcal{T}_X$, it holds that $\mathcal{B}(X) \subseteq \mathcal{G}$. By the definition of $\mathcal{G}$, this means that $A \cap E \in \mathcal{B}(E)$ for every $A \in \mathcal{B}(X)$, that is, $\mathcal{B}(X)|_E \subseteq \mathcal{B}(E)$.
Conclusion. Parts 3 and 4 give $\mathcal{B}(E) = \mathcal{B}(X)|_E$.

Remark (Discussion). Part (a) does not apply directly here. The set E need not be a Borel subset of X, so the family $\{A \in \mathcal{B}(X) : A \subseteq E\}$ of part (a) may not contain E. The trace is defined by intersecting with E, which avoids this.

4. (Rational half-lines generate the Borel σ-algebra)

4.a

Remark (Solution). Let $\mathcal{C}_1 := \{(-\infty, b] : b \in \mathbb{Q}\}$ and $\mathcal{C}_2 := \{(-\infty, b) : b \in \mathbb{Q}\}$. We show that $\sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2) = \mathcal{B}(\mathbb{R})$.
Step 1: $\sigma(\mathcal{C}_1), \sigma(\mathcal{C}_2) \subseteq \mathcal{B}(\mathbb{R})$**.** We show that $\mathcal{C}_1 \cup \mathcal{C}_2 \subseteq \mathcal{B}(\mathbb{R})$. This suffices because $\mathcal{B}(\mathbb{R})$ is a σ-algebra. Each $(-\infty, b)$ is open, so it belongs to $\mathcal{B}(\mathbb{R}) = \sigma(\text{open sets})$. Each $(-\infty, b] = \mathbb{R} \setminus (b, \infty)$ is the complement of an open set, which belongs to $\mathcal{B}(\mathbb{R})$, so $(-\infty, b] \in \mathcal{B}(\mathbb{R})$ because $\mathcal{B}(\mathbb{R})$ is closed under complements. Hence $\mathcal{C}_1 \cup \mathcal{C}_2 \subseteq \mathcal{B}(\mathbb{R})$. Then $\sigma(\mathcal{C}_1) \subseteq \mathcal{B}(\mathbb{R})$ and $\sigma(\mathcal{C}_2) \subseteq \mathcal{B}(\mathbb{R})$, because $\sigma(\mathcal{C}_i)$ is the smallest σ-algebra containing $\mathcal{C}_i$.
Step 2: $\sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2)$**.** We show that $\mathcal{C}_2 \subseteq \sigma(\mathcal{C}_1)$ and $\mathcal{C}_1 \subseteq \sigma(\mathcal{C}_2)$. This suffices because a σ-algebra containing a family contains the σ-algebra generated by it. Then $\sigma(\mathcal{C}_2) \subseteq \sigma(\mathcal{C}_1)$ and $\sigma(\mathcal{C}_1) \subseteq \sigma(\mathcal{C}_2)$. For $b \in \mathbb{Q}$,

$$(-\infty, b) = \bigcup_{n \geq 1} (-\infty, b - \tfrac{1}{n}], \qquad (-\infty, b] = \bigcap_{n \geq 1} (-\infty, b + \tfrac{1}{n}),$$

and $b \pm \frac{1}{n} \in \mathbb{Q}$. Here the index n runs over the integers $n \geq 1$. The first identity holds because $x < b$ implies $x \leq b - \frac{1}{n}$ for every integer $n \geq 1$ with $\frac{1}{n} \leq b - x$, and $x \leq b - \frac{1}{n}$ implies $x < b$. The second holds because $x \leq b$ implies $x < b + \frac{1}{n}$ for every $n \geq 1$, and $x < b + \frac{1}{n}$ for all $n \geq 1$ implies $x \leq b$. Thus $(-\infty, b)$ is a countable union of members of $\mathcal{C}_1$. Since $\sigma(\mathcal{C}_1)$ is closed under countable unions, $(-\infty, b) \in \sigma(\mathcal{C}_1)$. Hence $\mathcal{C}_2 \subseteq \sigma(\mathcal{C}_1)$. Also, $(-\infty, b]$ is a countable intersection of members of $\mathcal{C}_2$. Since $\sigma(\mathcal{C}_2)$ is closed under complements and countable unions, it is closed under countable intersections, so $(-\infty, b] \in \sigma(\mathcal{C}_2)$. Hence $\mathcal{C}_1 \subseteq \sigma(\mathcal{C}_2)$.
Step 3: $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\mathcal{C}_1)$**.** As shown in class (rational open cubes generate the Euclidean Borel sets), $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{R})$ with $\mathcal{R} := \{(p, q) : p, q \in \mathbb{Q},\ p < q\}$. We show that $\mathcal{R} \subseteq \sigma(\mathcal{C}_1)$, that is, $(p, q) \in \sigma(\mathcal{C}_1)$ for all rationals $p < q$. This suffices because $\sigma(\mathcal{C}_1)$ is a σ-algebra containing $\mathcal{R}$, and $\sigma(\mathcal{R})$ is the smallest such σ-algebra. Hence $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{R}) \subseteq \sigma(\mathcal{C}_1)$. Let $p < q$ be rational. Then $(p, q) = (-\infty, q) \setminus (-\infty, p] = (-\infty, q) \cap (\mathbb{R} \setminus (-\infty, p])$. Here $(-\infty, q) \in \mathcal{C}_2 \subseteq \sigma(\mathcal{C}_2) = \sigma(\mathcal{C}_1)$ by Step 2, and $(-\infty, p] \in \mathcal{C}_1 \subseteq \sigma(\mathcal{C}_1)$. Since $\sigma(\mathcal{C}_1)$ is closed under complements and finite intersections, $(p, q) \in \sigma(\mathcal{C}_1)$.
Together with Steps 1 and 2, $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2) \subseteq \mathcal{B}(\mathbb{R})$.

5. (Sections of product measurable sets)

5.a

Remark (Solution). For $S \subseteq X \times Y$ the sections are $S^y \coloneqq \{x \in X : (x,y) \in S\}$ and $S_x \coloneqq \{y \in Y : (x,y) \in S\}$. We show that for $S \in \mathcal{E} \otimes \mathcal{F}$, $S^y \in \mathcal{E}$ for every $y \in Y$ and $S_x \in \mathcal{F}$ for every $x \in X$.
Sections in the first coordinate. Fix $y \in Y$ and let $\mathcal{G}^y \coloneqq \{S \subseteq X \times Y : S^y \in \mathcal{E}\}$. We show that $\mathcal{G}^y$ is a σ-algebra on $X \times Y$ containing every measurable rectangle; then $\mathcal{E} \otimes \mathcal{F} \subseteq \mathcal{G}^y$.

- $\mathcal{G}^y$ contains every measurable rectangle. Indeed, for $A \in \mathcal{E}$ and $B \in \mathcal{F}$, $(A \times B)^y = A$ if $y \in B$ and $(A \times B)^y = \emptyset$ otherwise. Both A and $\emptyset$ belong to $\mathcal{E}$.
- $\mathcal{G}^y$ is a σ-algebra on $X \times Y$. Directly from the definition of the section, $(X \times Y)^y = X$, $\big((X \times Y) \setminus S\big)^y = X \setminus S^y$ and $\big(\bigcup_n S_n\big)^y = \bigcup_n S_n^y$. Since $\mathcal{E}$ is a σ-algebra on X, it contains X and is closed under complements and countable unions. Hence $X \times Y \in \mathcal{G}^y$. If $S \in \mathcal{G}^y$, then $S^y \in \mathcal{E}$, so $\big((X \times Y) \setminus S\big)^y = X \setminus S^y \in \mathcal{E}$ and $(X \times Y) \setminus S \in \mathcal{G}^y$. If $S_n \in \mathcal{G}^y$ for $n \in \mathbb{N}$, then $S_n^y \in \mathcal{E}$ for every n, so $\big(\bigcup_n S_n\big)^y = \bigcup_n S_n^y \in \mathcal{E}$ and $\bigcup_n S_n \in \mathcal{G}^y$.

Hence $\mathcal{G}^y$ is a σ-algebra containing the family of measurable rectangles. Since $\mathcal{E} \otimes \mathcal{F} = \sigma(\text{rectangles})$ is the smallest σ-algebra containing that family, $\mathcal{E} \otimes \mathcal{F} \subseteq \mathcal{G}^y$. The argument used no property of y other than $y \in Y$. Therefore $S^y \in \mathcal{E}$ for every $S \in \mathcal{E} \otimes \mathcal{F}$ and every $y \in Y$.
Sections in the second coordinate. The argument for the sections S_x is, mutatis mutandis, the same. Fix $x \in X$ and let $\mathcal{G}_x \coloneqq \{S \subseteq X \times Y : S_x \in \mathcal{F}\}$. The roles of X, $\mathcal{E}$ and y are exchanged with those of Y, $\mathcal{F}$ and x, and for a measurable rectangle $(A \times B)_x = B$ if $x \in A$ and $(A \times B)_x = \emptyset$ otherwise. Therefore $S_x \in \mathcal{F}$ for every $S \in \mathcal{E} \otimes \mathcal{F}$ and every $x \in X$.